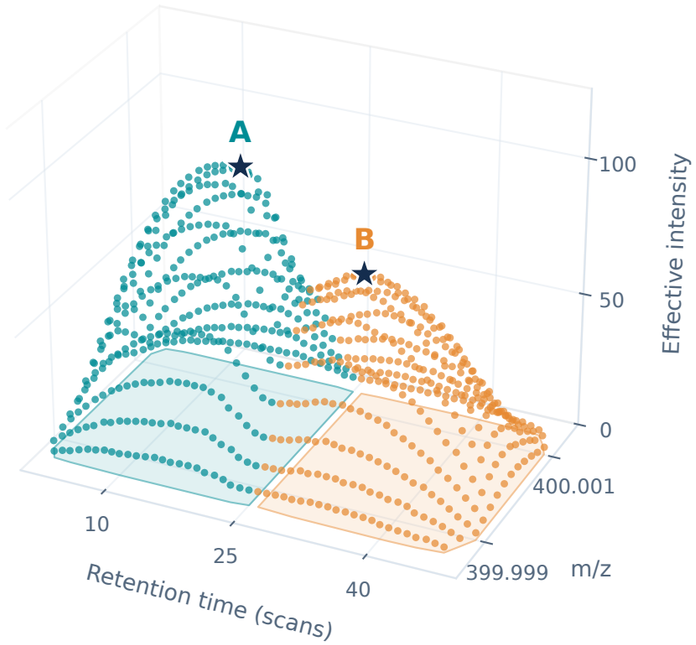
A shallow split becomes one basin

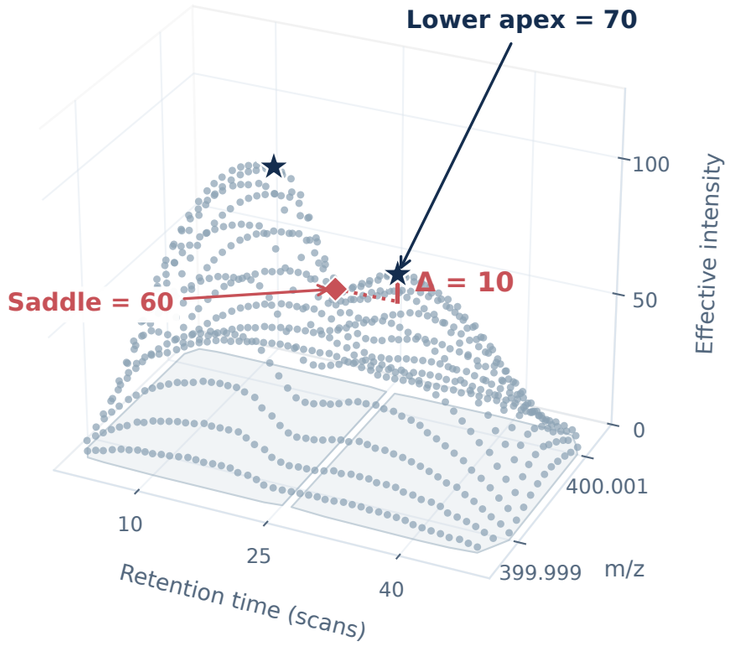
Compare the lower apex with the saddle connecting the two initial basins.

1 Two initial basins



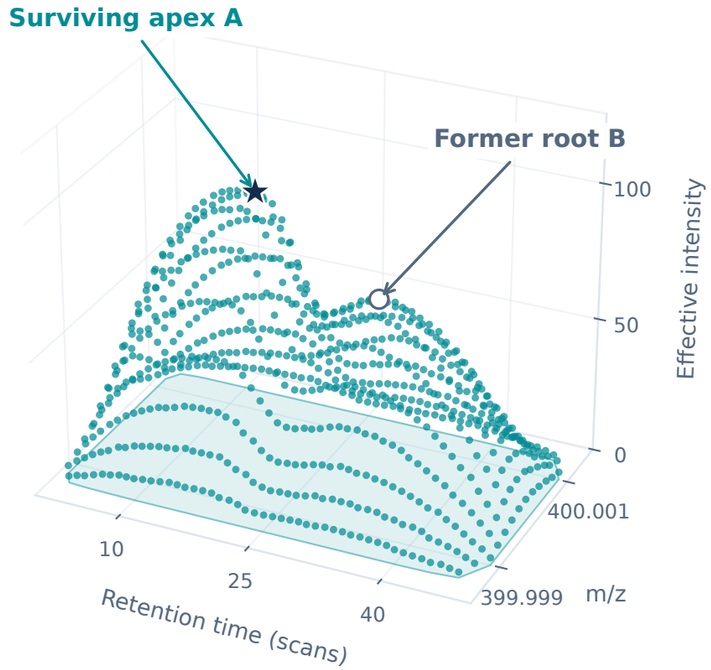
Each coloured cloud has its own apex.
The basins meet at a connecting saddle.

2 Measure the connecting dip



Persistence = 70 − 60 = 10.
10 < threshold 35 → MERGE.

3 Merge into basin A



All points now share basin A.
The lower maximum remains in the data.

Here: threshold = max(0.5 × lower apex, 3 × local noise) = 35. Apex spacing = 1.4 median widths.

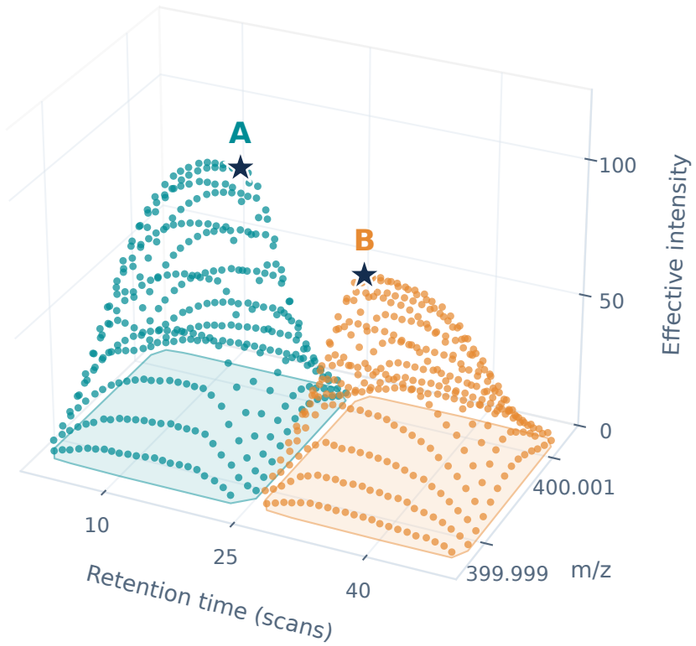
Merging changes basin membership. The point cloud itself stays unchanged.

Synthetic point clouds • actual Peak3D decisions • default fractional rule shown; spacing and noise also affect merging

A deep valley keeps two basins separate

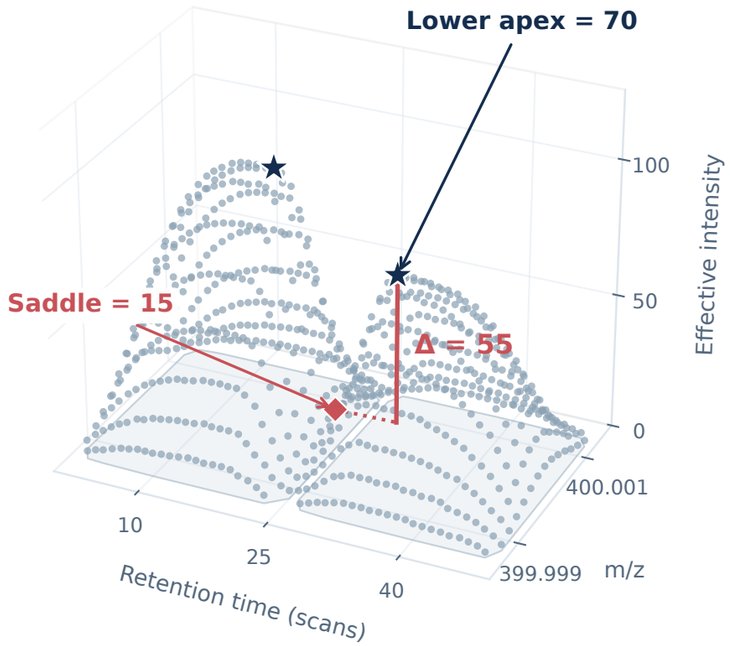
Compare the lower apex with the saddle connecting the two initial basins.

1 Two initial basins



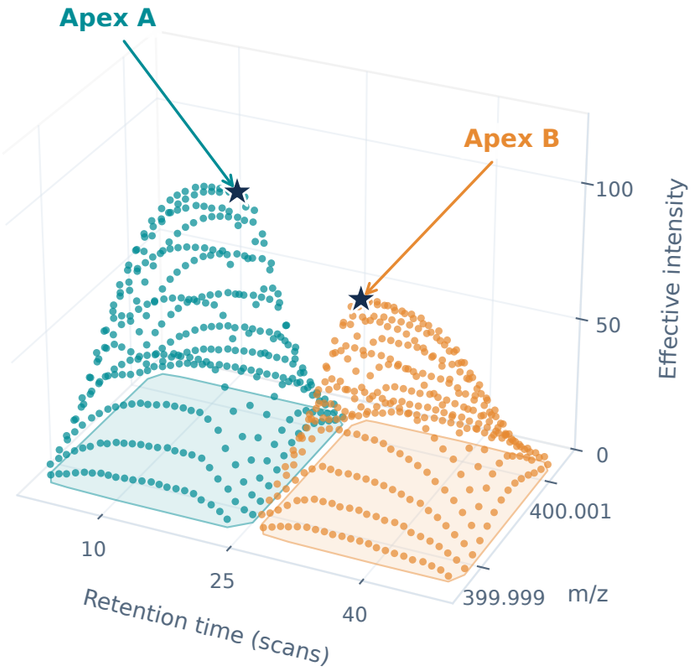
Each coloured cloud has its own apex.
The basins meet at a connecting saddle.

2 Measure the connecting dip



Persistence = 70 − 15 = 55.
55 ≥ threshold 35 → KEEP SEPARATE.

3 Keep basins A and B



Two basins and two apexes survive.
Their memberships remain separate.

Here: threshold = max(0.5 × lower apex, 3 × local noise) = 35. Apex spacing = 1.4 median widths.

The lower apex clears the persistence threshold, so both basins retain their own apex.

Synthetic point clouds • actual Peak3D decisions • default fractional rule shown; spacing and noise also affect merging